Coding OTS RL Envs

546	16%	2%
Tasks	Gemini 4 Argon pass rate	Gemini 3.8 Flash pass rate

We selected coding RL Envs that challenge Gemini models and grouped them into four catalogs.

Catalog	Tasks	What the tasks do	% of all tasks	Average turns, Gemini 4 Argon	Average turns, Gemini 3.8 Flash
ProgramBench and Migration	281	Rebuild or upgrade programs to match a reference exactly, or port libraries to another language.	51.5%	426	440
System design & IaC	75	Keep running services and live infrastructure correct through failures and refactors.	13.7%	277	236
Diagnosis, Debugging, and Bug Fixing	82	Find and fix faults in real codebases from symptoms or documentation.	15.0%	262	534
Feature implementation	108	Add capabilities to existing libraries and services without breaking current behavior.	19.8%	232	361
Total	**546**		**100%**		

Samples

Each sample is a full task. Click its task id or the download button to download it. Each main sample's prompt and environment diagram are on its catalog page in the full report. Attached samples are extra tasks shared as downloads only.

Sample	Catalog	Example	Task id	Gemini 4 Argon	Gemini 3.8 Flash
Main	ProgramBench and Migration	**Example 1: Statistics & numerical methods (Go)**	`dynahist-go-port` ↓download	33%	0%
Main	ProgramBench and Migration	**Example 2: Media processing (Agent's choice)**	`svtav1-encoder-rebuild` ↓download	0%	
Main	ProgramBench and Migration	**Example 3: Graph algorithms (Go)**	`dagre-go-port` ↓download	50%	0%
Main	System design & IaC	**Example 1: Payments & banking (TypeScript)**	`actual-sync-compaction` ↓download	20%	
Main	System design & IaC	**Example 2: Secrets & access control (Go)**	`harbor-scanner-credentials` ↓download	20%	
Main	System design & IaC	**Example 3: Software supply chain (C#)**	`bagetter-fenced-writes` ↓download	20%	

Main	Diagnosis, Debugging, and Bug Fixing	**Example 1: Compilers & language runtimes (Python)**	`exo-canonical-ots` ↓download	0%	
Main	Diagnosis, Debugging, and Bug Fixing	**Example 2: Medical imaging & healthcare (Python)**	`torchio-canonical-ots` ↓download	0%	
Main	Diagnosis, Debugging, and Bug Fixing	**Example 3: Machine learning & MLOps (Python)**	`swanlab-nanpair-ots` ↓download	0%	
Main	Feature implementation	**Example 1: Databases & storage engines (Go)**	`memdb-txn-ots` ↓download	67%	
Main	Feature implementation	**Example 2: Databases & storage engines (Python)**	`sqlite-history-repair` ↓download	0%	60%
Main	Feature implementation	**Example 3: Data pipelines & ETL (Python)**	`etlplus-csv-pipeline` ↓download	0%	100%
Attached	ProgramBench and Migration	Probabilistic data structures (Go)	`heavykeeper-go-port` ↓download	75%	
Attached	System design & IaC	Cloud infra: containers & self-hosted platforms (HCL (Terraform))	`dualipv6-module-refactor` ↓download	0%	0%
Attached	System design & IaC	Cloud infra: AWS platform stacks (HCL (Terraform))	`centralwatch-module-refactor` ↓download	0%	100%
Attached	System design & IaC	Cloud infra: AWS platform stacks (HCL (Terraform))	`modelrouter-module-refactor` ↓download	50%	100%
Attached	System design & IaC	Cloud infra: containers & self-hosted platforms (TypeScript)	`atlasgrid-pulumi-migration` ↓download	50%	67%
Attached	System design & IaC	Web applications (TypeScript)	`dittofeed-user-deletion` ↓download	75%	
Attached	System design & IaC	Media processing (TypeScript)	`immich-storage-quotas` ↓download	80%	
Attached	System design & IaC	Payments & banking (TypeScript)	`medusa-cart-completion` ↓download	80%	
Attached	Diagnosis, Debugging, and Bug Fixing	Web crawling (Python)	`scrapy-pdk-ots` ↓download	67%	100%
Attached	Feature implementation	Data formats & serialization (Python)	`xmltodict-nulls-etl` ↓download	50%	100%

Where Gemini 4 Argon fails

Across the 22 samples, Gemini 4 Argon did not pass 39 of its 65 graded runs and passed the other 26. We sorted the failures into five failure modes. Each sample is counted once, under its main failure mode, though single runs can fail in different ways.

39 of 65
Argon runs not passed

33%
Argon pass@1 on samples

57%
Argon pass@2 on the 21 samples with 2+ runs

Failure mode	Details and examples	Runs not passed	Samples
Workflow and access failures	**Missed cleanup, conflict handling or permitted access during a workflow.** • **Cancelled upload:** Cancelling an upload should free its reserved space. Argon's version kept the space occupied, so the next upload was rejected as over quota. • **Overlapping upload and delete:** A valid package upload should still finish when a delete overlaps with it. Argon's version rejected the upload as a conflict. • **Download permissions:** A scanner should be allowed to download from its assigned repository. Argon's security change blocked that permitted download.	15	6
Incorrect data or state	**Lost values, changed future behavior or missed database-change notifications.** • **Saving and reopening a frequency counter:** After reopening, the counter used a different random sequence, so it no longer continued identically when given the same new inputs. • **Adding CSV support:** The data-processing tool should preserve correct summary totals. Argon's changes produced a count of 75 matching records where 119 were expected. • **Restoring an older database:** Restoration should preserve values in columns added later. Argon's version lost those values when restoring an older-format database.	6	4
Incomplete ports and rebuilds	**Failed to preserve the original behavior across inputs or deliver a program that builds.** • **Porting a counting library:** Ordinary counts worked, but the largest supported count failed in one storage format. The port should match the original at that limit. • **Rebuilding a video encoder:** Argon selected from fixed encoded samples, which could not handle the wider range of video inputs. Only 32 of 453 checks passed. • **Porting a graph library:** Duplicate functions and incompatible function calls stopped the submitted code from compiling, so none of the behavior checks could run.	5	3
Missing required behavior	**Passed many checks while missing required modes, interfaces or processing paths.** • **Combining image filters:** Argon's repair still failed when inclusion and exclusion filters were used together. Only images allowed by both rules should be processed. • **Fixing incorrect experiment results:** Failed training runs were being reported as zero loss. Argon reproduced the bug, but its fix covered only two of four tested variants.	6	4
Unfinished integration and features	**Left configuration dependencies, module connections or requested API options incomplete.** • **Changing a deployment's project name:** Generated machine startup scripts still contained the old project name. The setting should have updated everywhere it was used. • **Splitting infrastructure into modules:** The top-level configuration still read outputs from the old resources, rather than from the new modules as required. • **Adding XML options:** New options were meant to control null values and whitespace. Argon's submitted library rejected both options as unrecognized arguments.	7	5
Total		**39**	**22**

Headroom for improvement

Across the 22 samples, the chance that one Argon run passes (pass@1) is 33%. On the 21 samples with two or more graded Argon runs, the chance that at least one of two runs passes (pass@2) is 57%. Each sample is weighted equally. Argon can often reach a working solution but does not get there reliably.

Argon often implements much of the solution, but misses conditions that matter in real use: cancelling an upload, restoring older data, combining settings, or keeping totals correct after a change. More training examples covering these situations can help strengthen that behavior. These environments provide concrete failure cases, realistic implementation work, and executable checks to measure whether the gaps are closing.

Pass rates by catalog

Each cell is a model's pass rate in one catalog: its passed runs over its graded runs there, with errored runs left out.

Pass rate by model and catalog

	ProgramBench and Migration	System design & IaC	Diagnosis, Debugging, and Bug Fixing	Feature implementation
google/gemini-4-argon	**13%** 117 tasks	**27%** 63 tasks	**11%** 17 tasks	**5%** 30 tasks
google/gemini-3.8-flash	**0%** 168 tasks	**16%** 20 tasks	**4%** 72 tasks	**2%** 99 tasks

Model	Pass rate
google/gemini-4-argon	16%
google/gemini-3.8-flash	2%

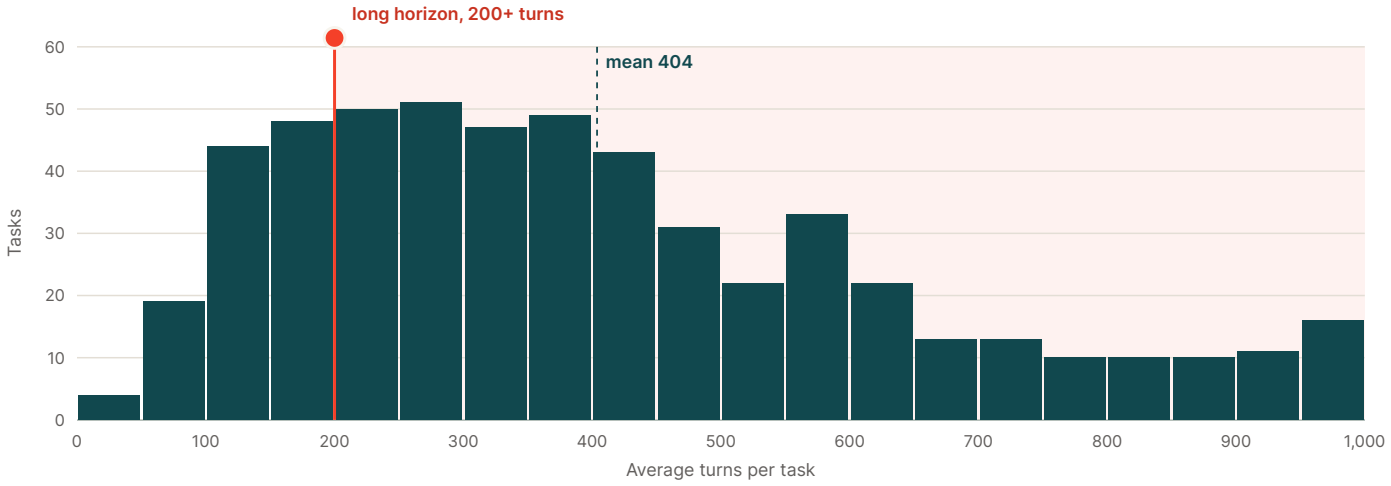
Turn counts

A task's turn count is the average over its runs by Gemini 4 Argon and Gemini 3.8 Flash. Across all catalogs the average is 404 turns, and 79% of tasks average 200 turns or more.

Turns per task, all catalogs

google/gemini-4-argon 227 tasks | google/gemini-3.8-flash 359 tasks

431 of 546 tasks (79%) average 200 turns or more. Mean 404 turns per task.



Long-horizon tasks (200+ turns) by catalog.

Catalog	Long-horizon tasks	Tasks with runs	Share
ProgramBench and Migration	239	281	85%
Feature implementation	77	108	71%
Diagnosis, Debugging, and Bug Fixing	64	82	78%
System design & IaC	51	75	68%